

Specialized — but flexible — prefrontal networks for auditory and visual cognition

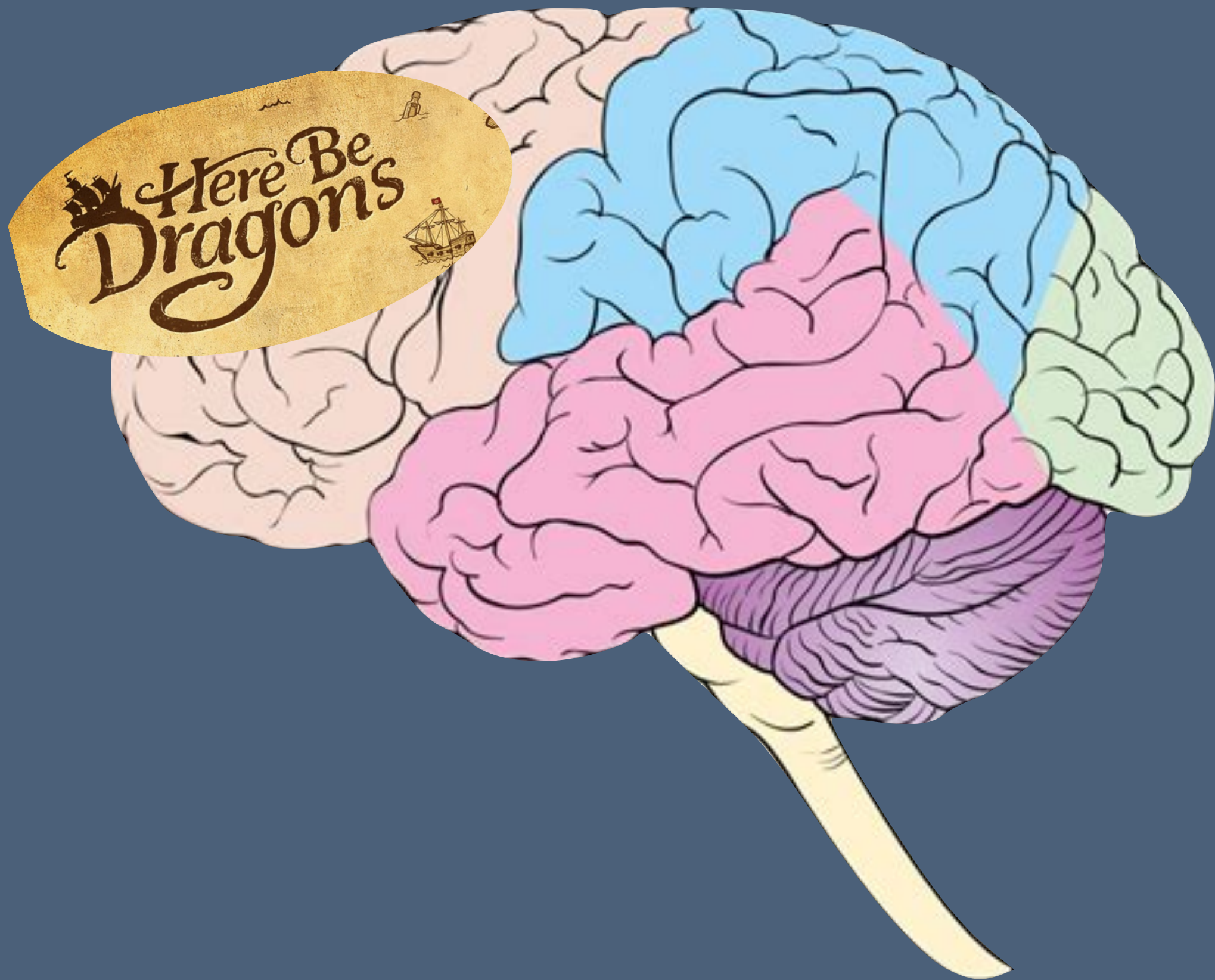
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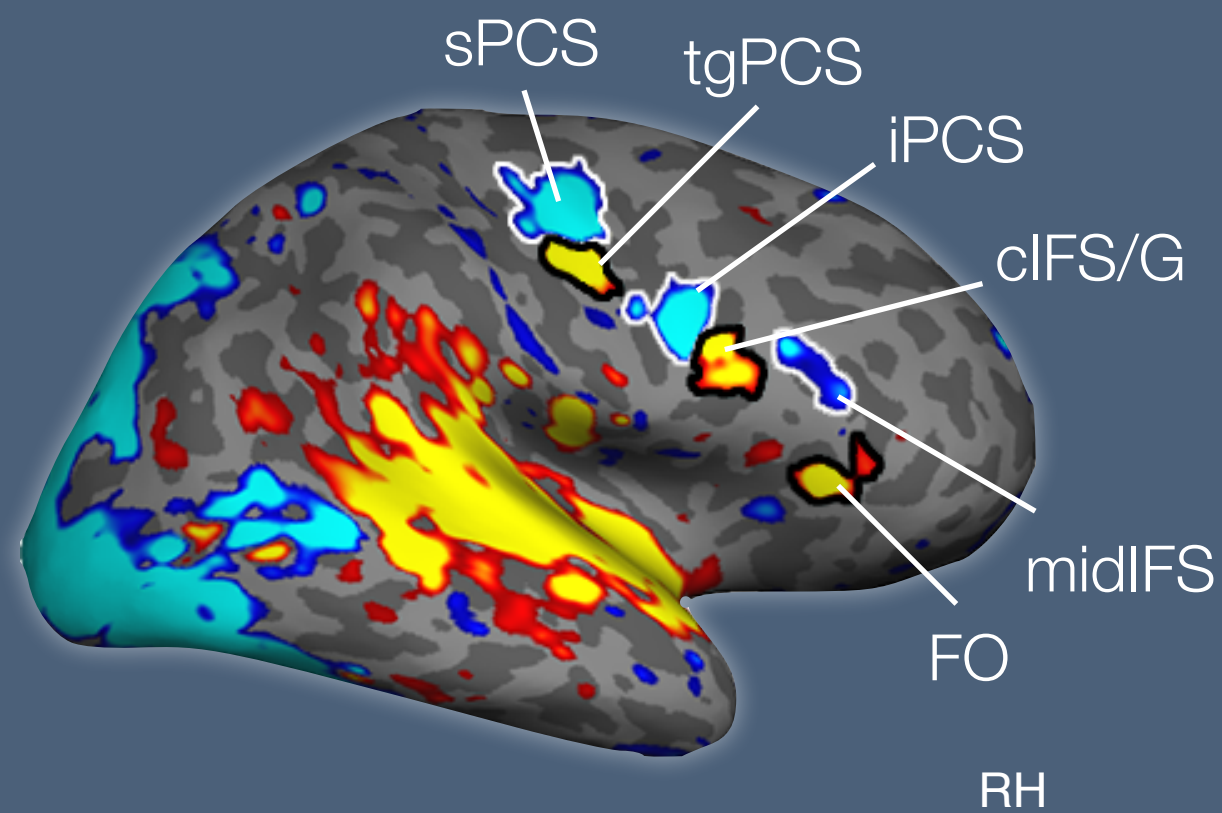
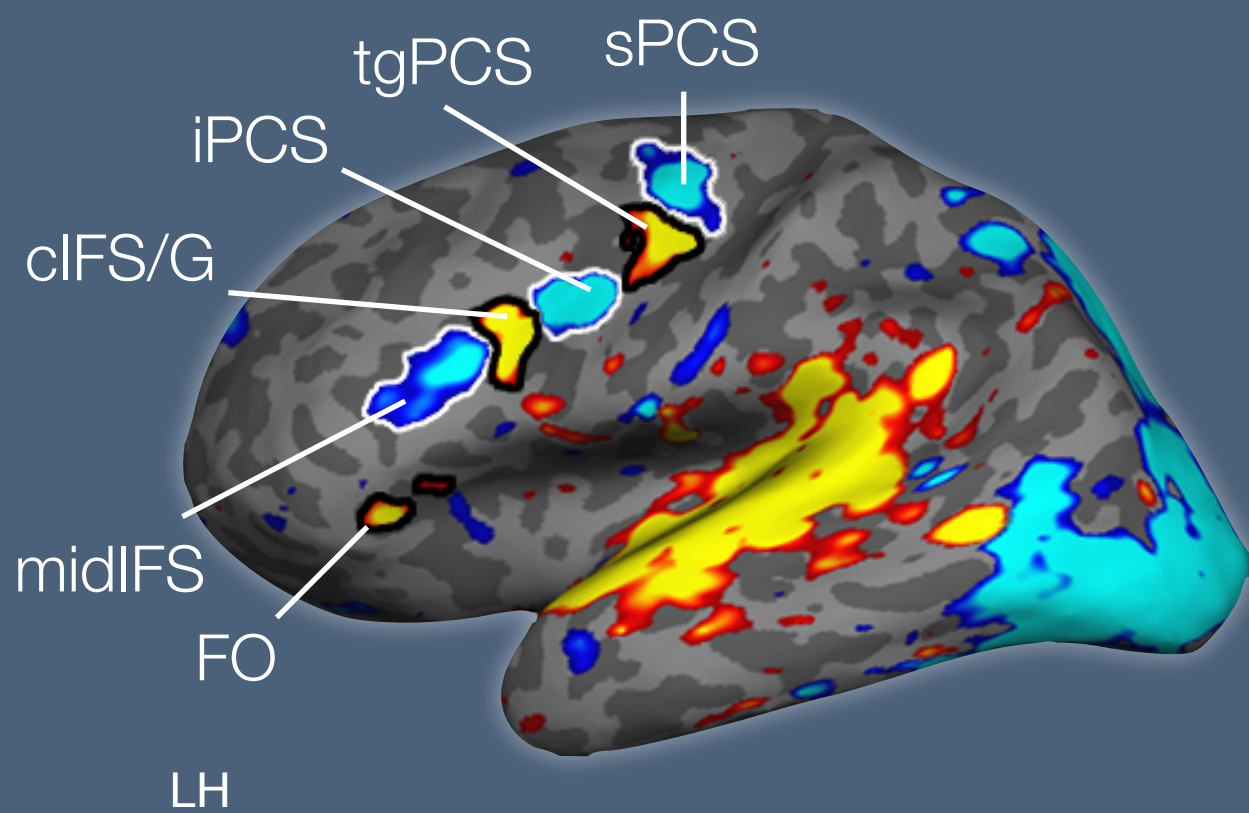
**Carnegie
Mellon
University**



All the references!



Localizing auditory and visual PFC networks in individual subjects



Visual

Auditory



5

1.3

1.3

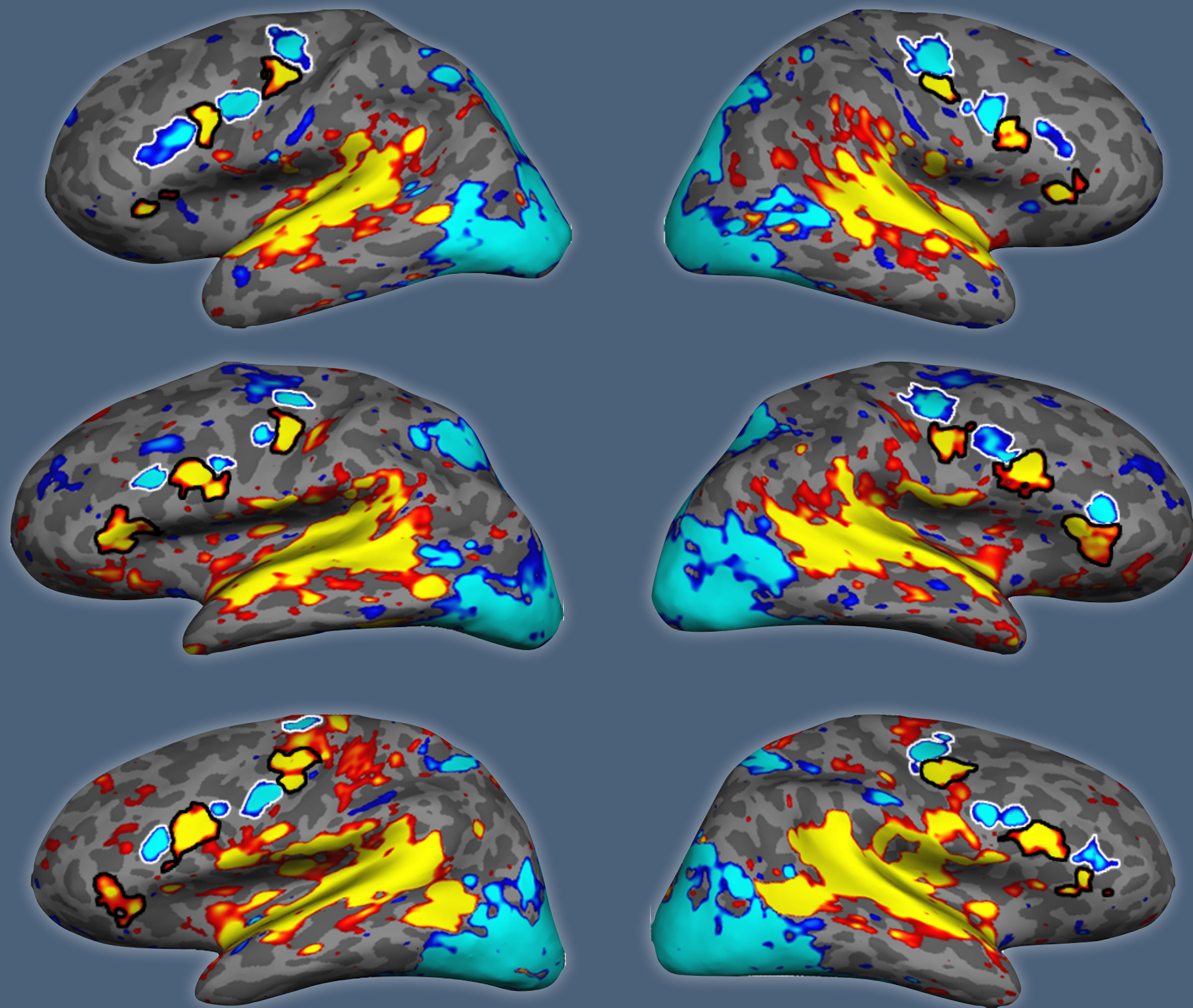
5

$-\log p$

Tobyne 2025 *JNeuro.*

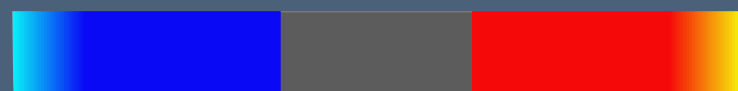
Noyce 2022 *CerCor.*

Noyce 2017 *JNeuro.*



Visual

Auditory



5

1.3

1.3

5

$-\log p$

Tobyne 2025 *J. Neuro.*

Noyce 2022 *Cer. Cor.*

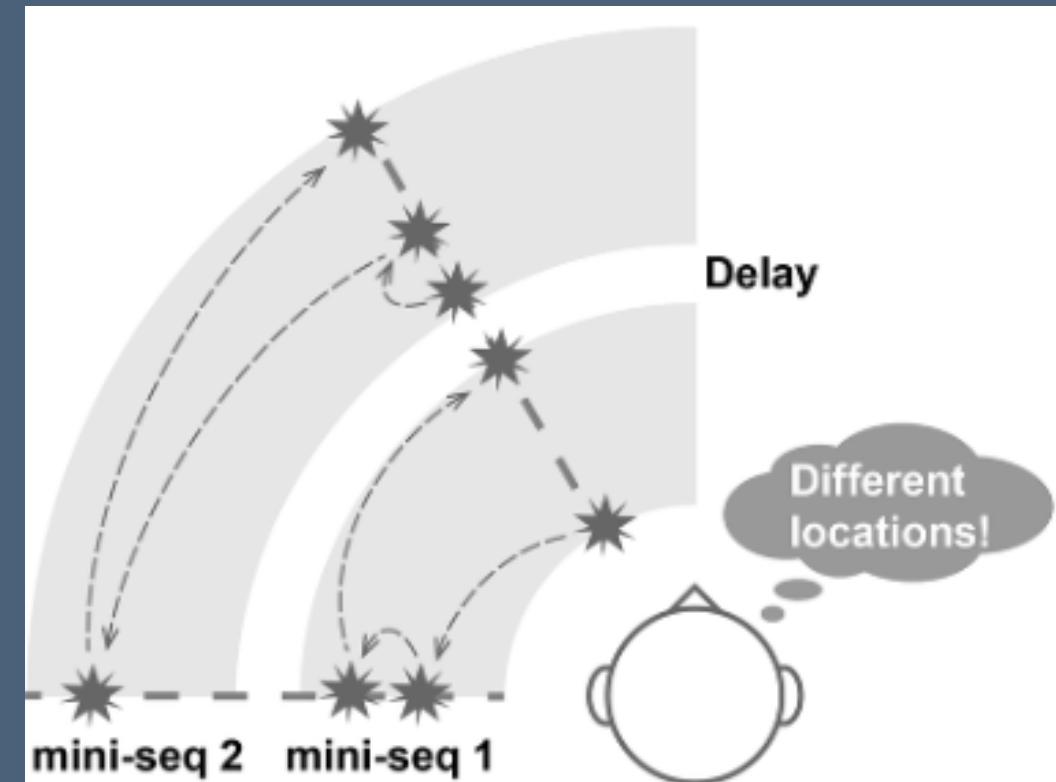
Noyce 2017 *J. Neuro.*

What are these sensory-biased PFC networks doing?

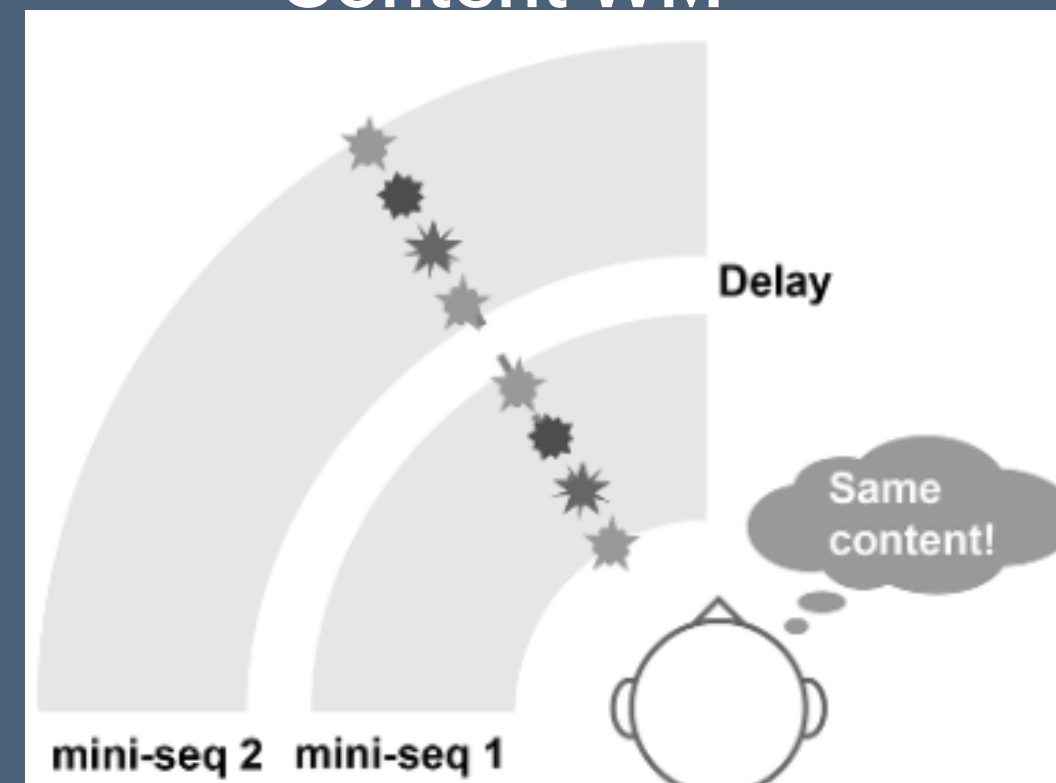
Cross-modal flexibility during auditory working memory

- Spatialized with **HRTFs**, **ILDs**, or **ITDs**.
- 3 tasks: **Spatial WM**, **Content WM**, and **Passive Listening**
- 2 hemifields: **Left**, **Right**

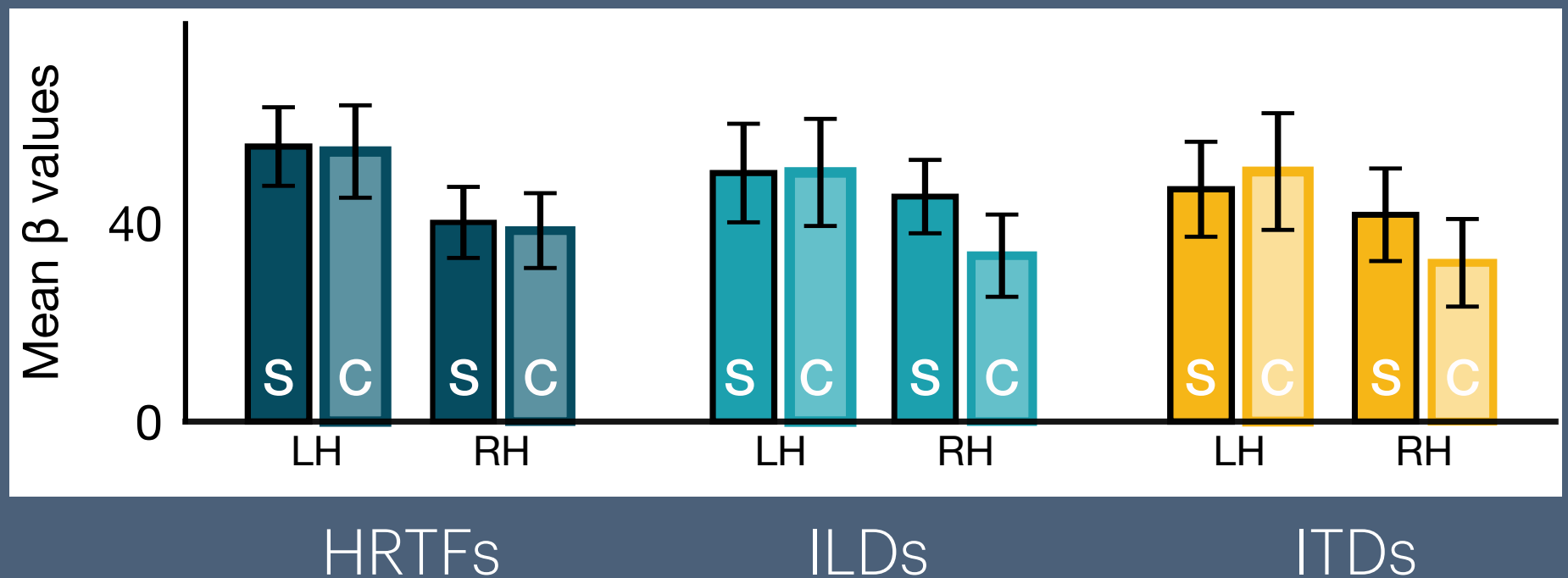
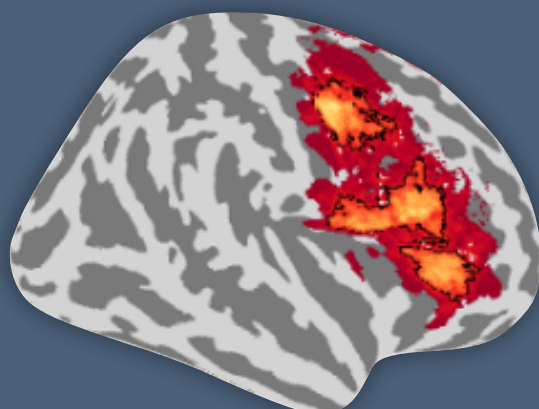
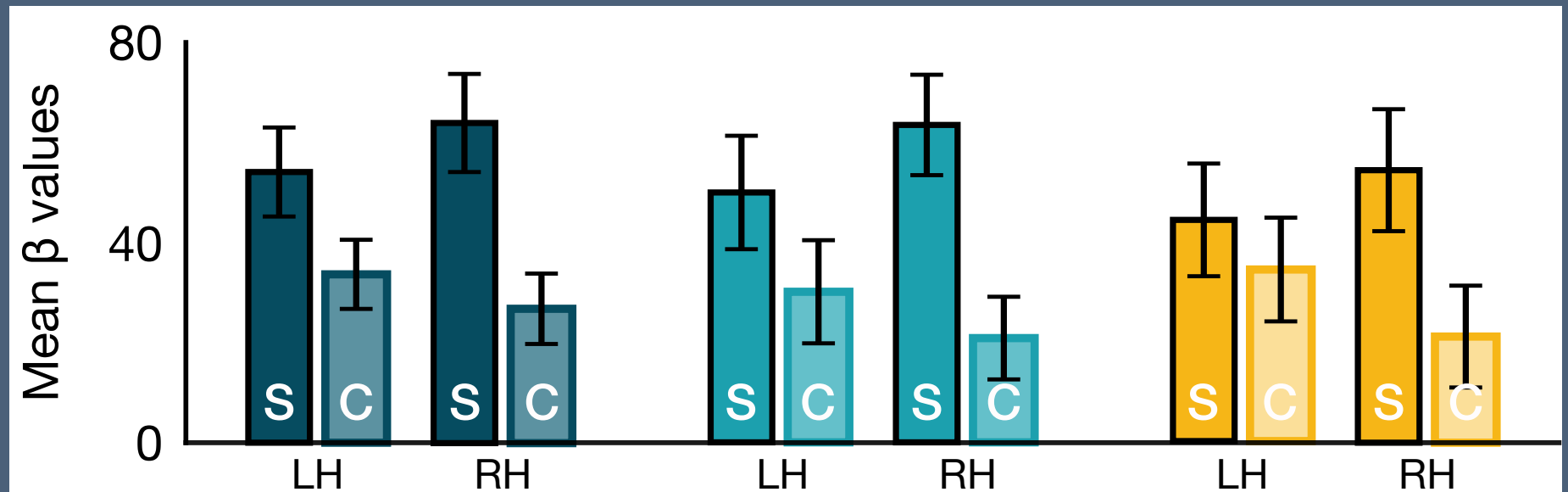
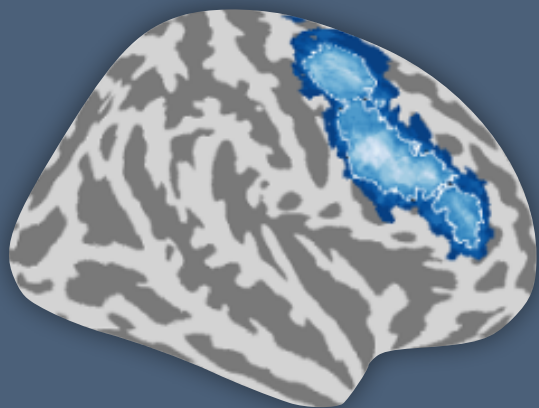
Spatial WM



Content WM



Evidence of cross-modal flexibility in auditory working memory

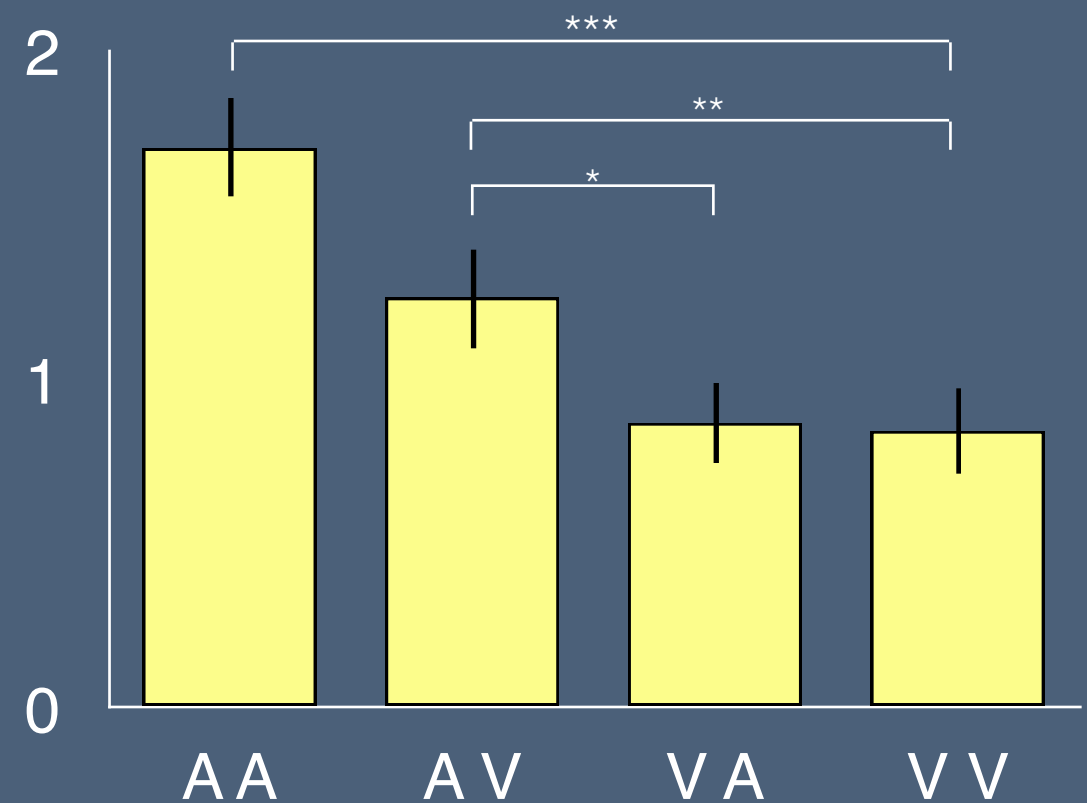


Cross-modal flexibility
depends on computational
needs

Visual, auditory, and crossmodal working memory



Space Memory



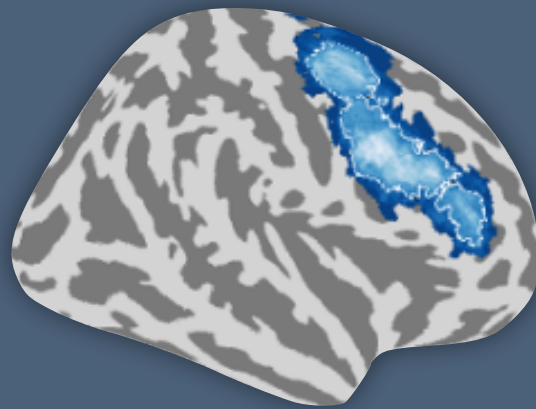
Time Memory

- Modality appropriateness effect.
- No crossmodal cost (AV and VA as good or better than the weaker unimodal condition).
- Task-appropriate modality at encoding is beneficial.

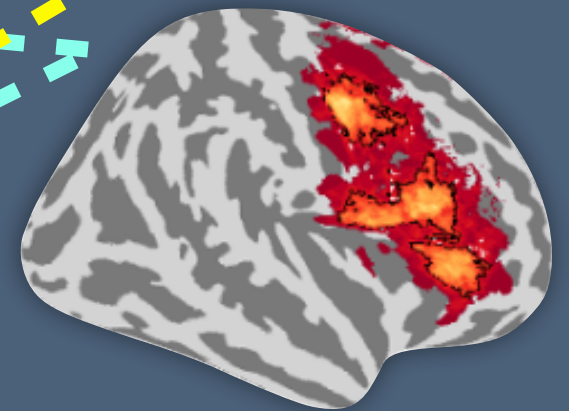
When does the brain bring in the “other” network?



Spatial features

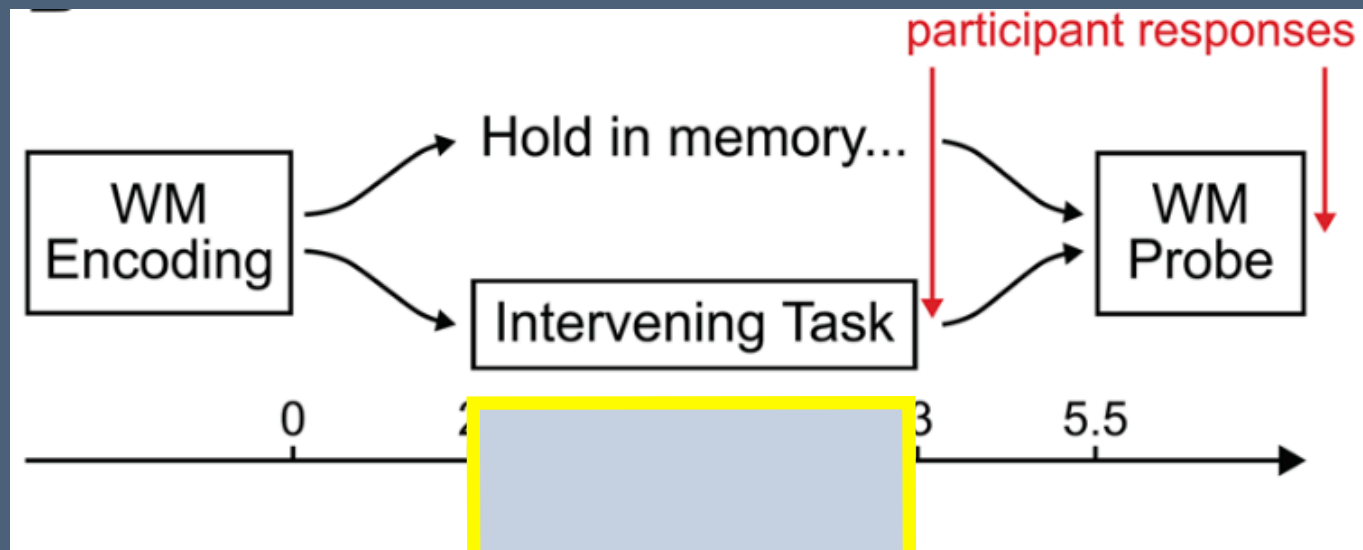


1. Encode + maintain features in memory
2. Explicitly compare senses
3. Deploy + control attention



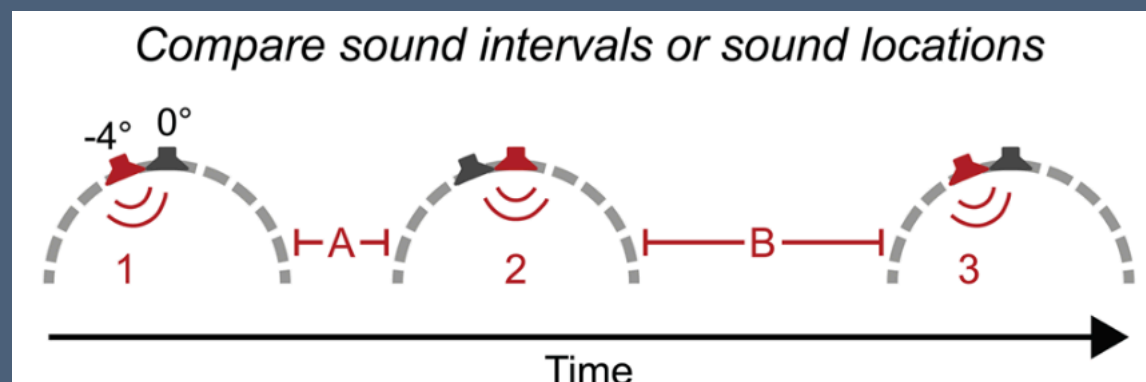
Spectral + Temporal features

Dual-task interference reveals shared network loading.



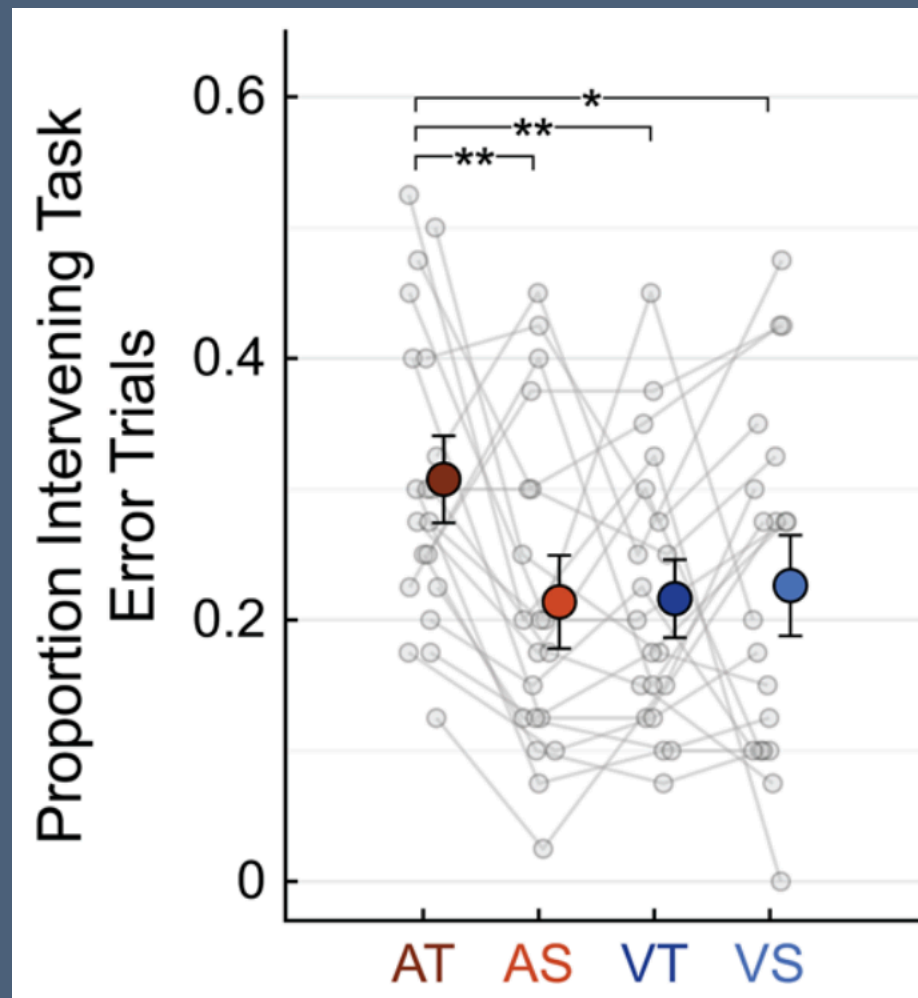
- **Hypothesis:** If the WM task and the intervening judgment load the same networks, then

- Poorer task performance
- Greater effort → pupil diameter
- Decreased neural processing → ERPs



Dual-task interference reveals shared network loading.

Intervening: Judge *timing*



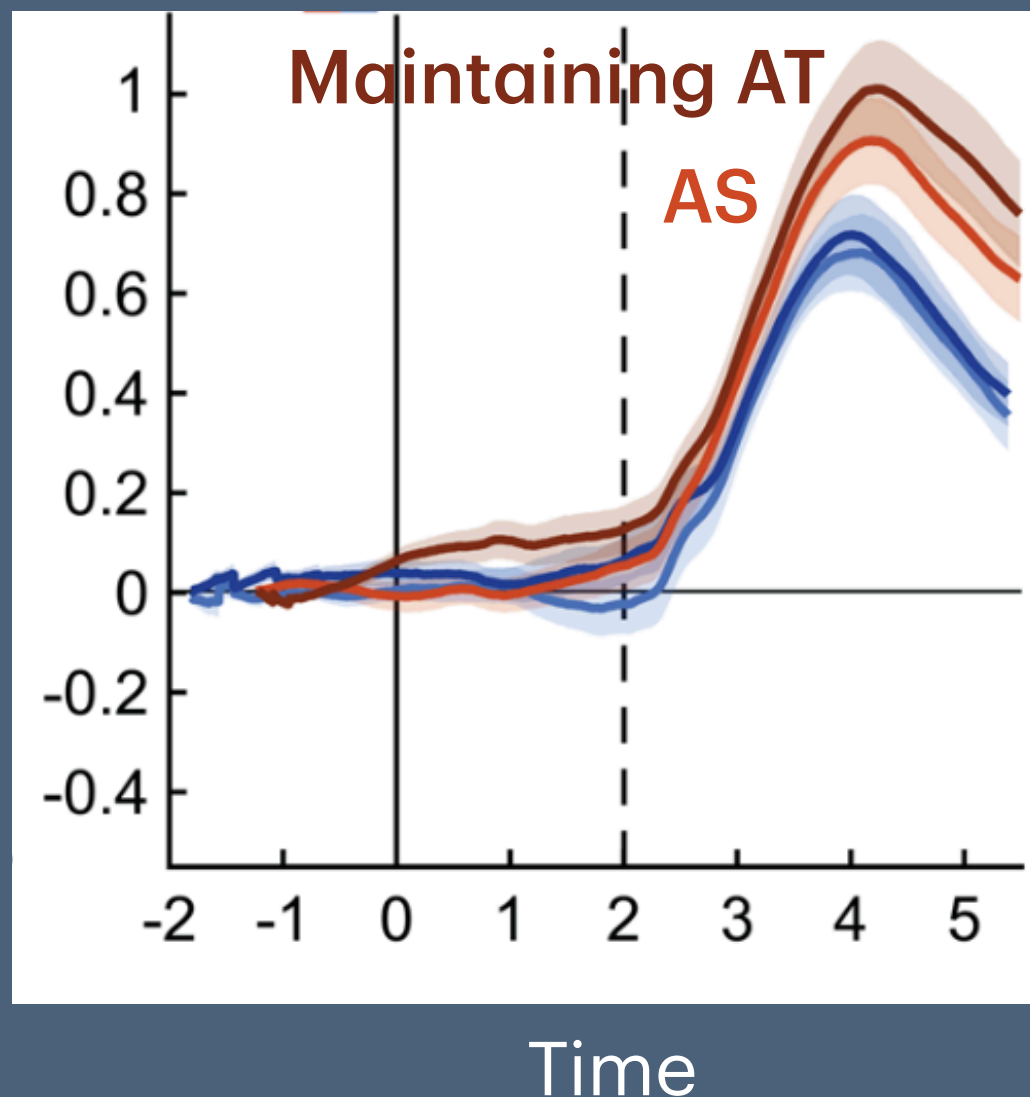
WM task conditions

- **Hypothesis:** If the WM task and the intervening judgment load the same networks, then
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Dual-task interference reveals shared network loading.

Intervening: Judge *timing*

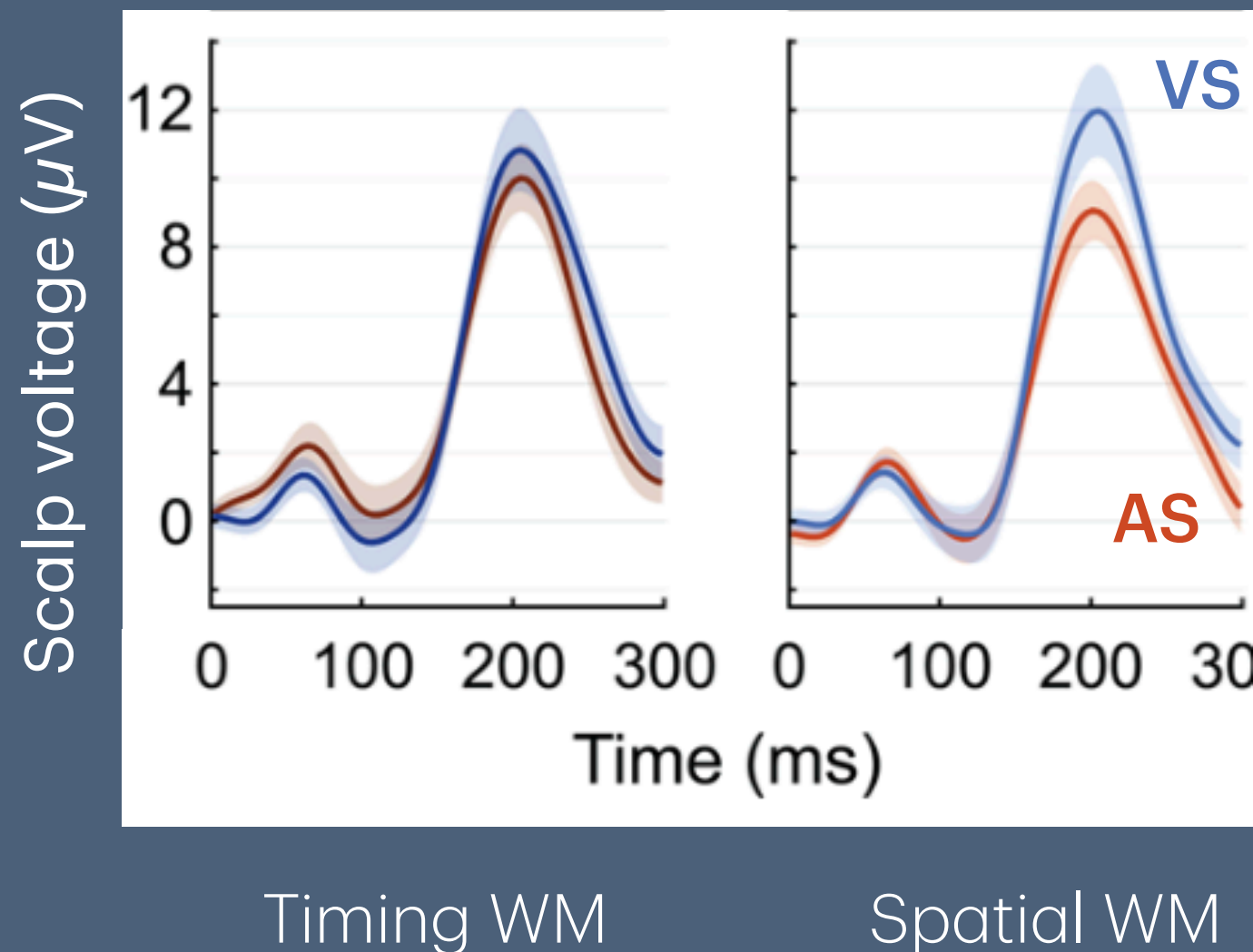
Pupil response to intervening task



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Dual-task interference reveals shared network loading.

Intervening: Judge *timing*



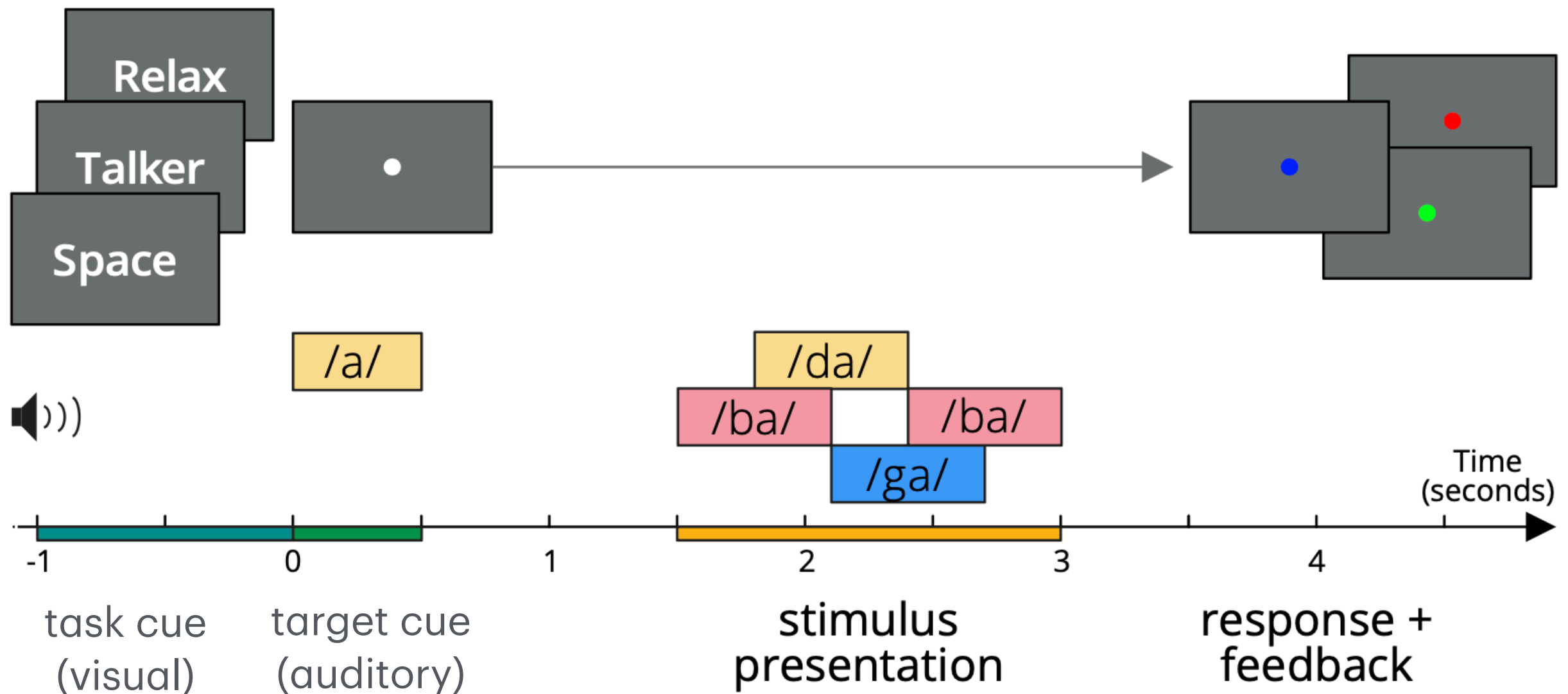
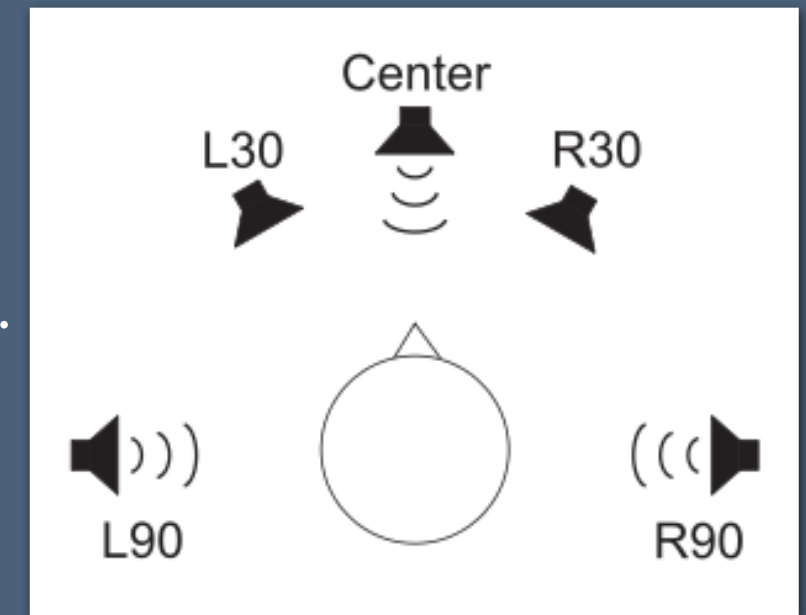
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Mapping the neural representation of attention

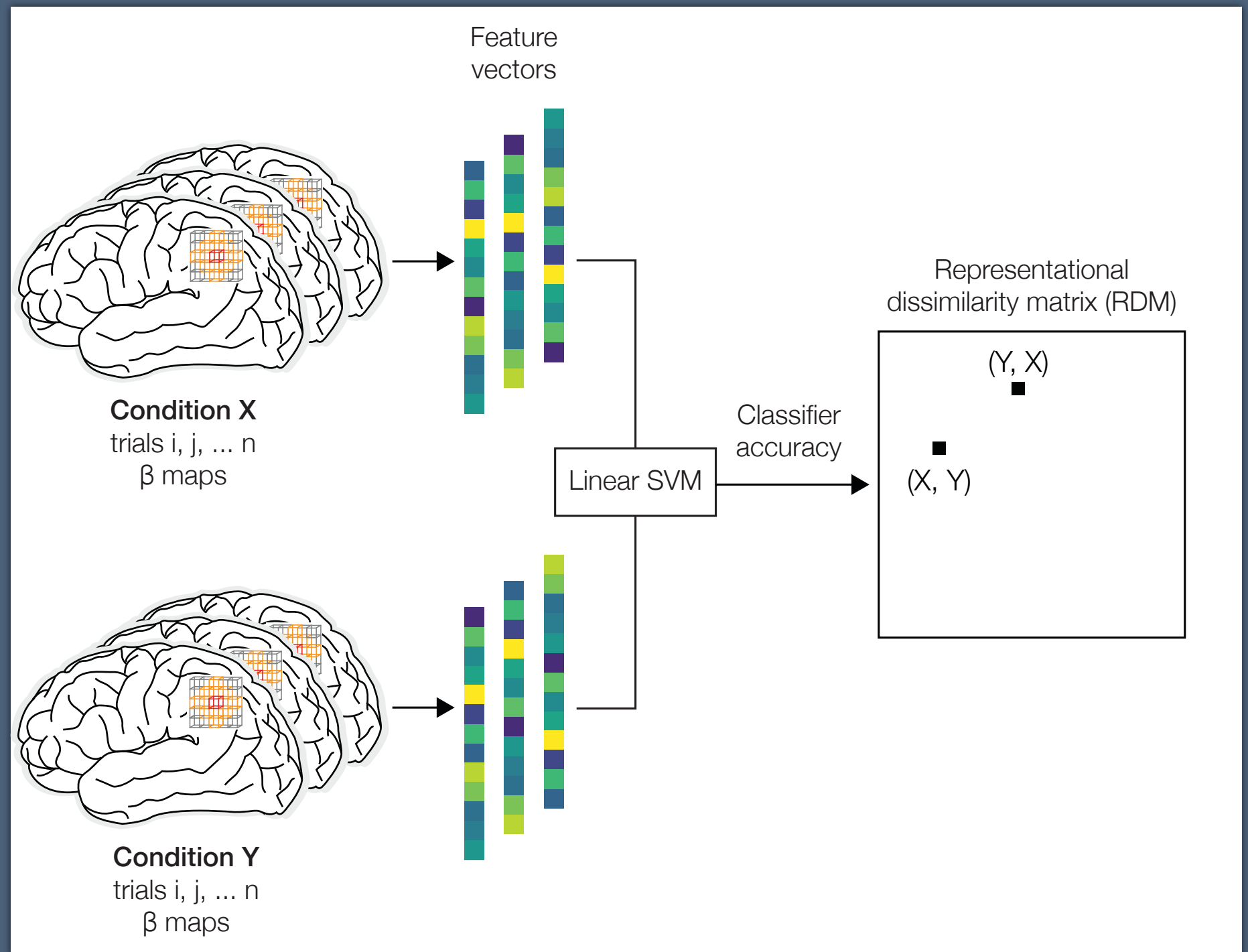
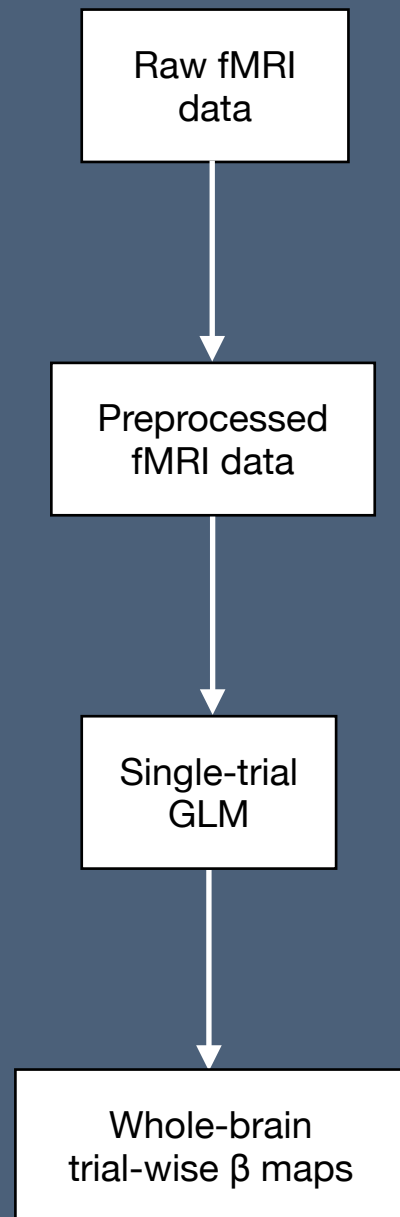
Extending representational similarity analysis to the study of attention

- If two conditions are similar (or dissimilar) in some feature space...
 - and the brain activity is similar (or dissimilar) between the two conditions...
 - then that feature is probably represented in this brain activity.
- Measuring pairwise dissimilarity across many conditions and comparing to hypothesized patterns.
- RSA has been widely used to study perceptual processing.
 - Vision (Cichy 2014), AV integration (Cecere 2015), speech (Lowe 2021), etc.

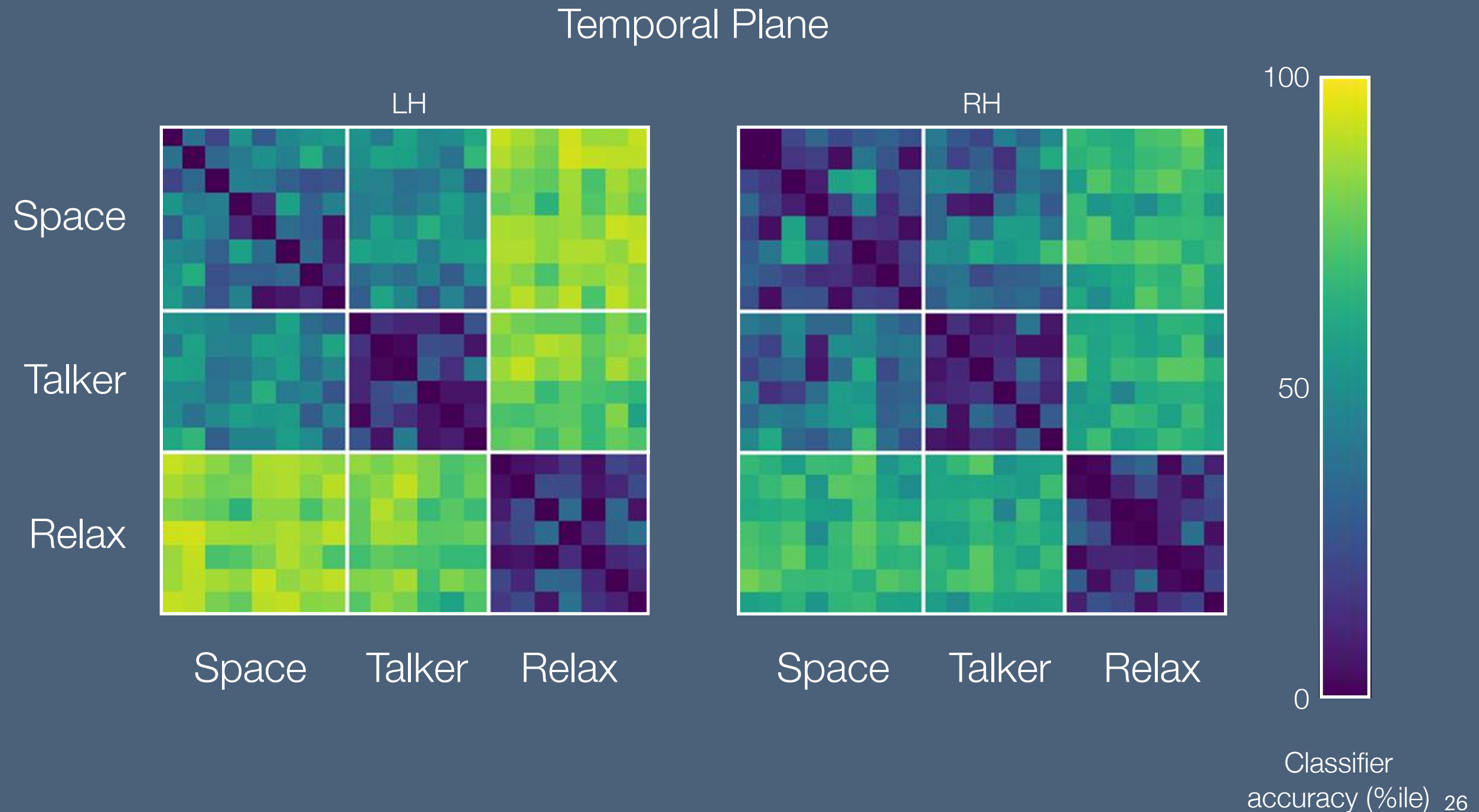
Condition-rich feature space for auditory attention.



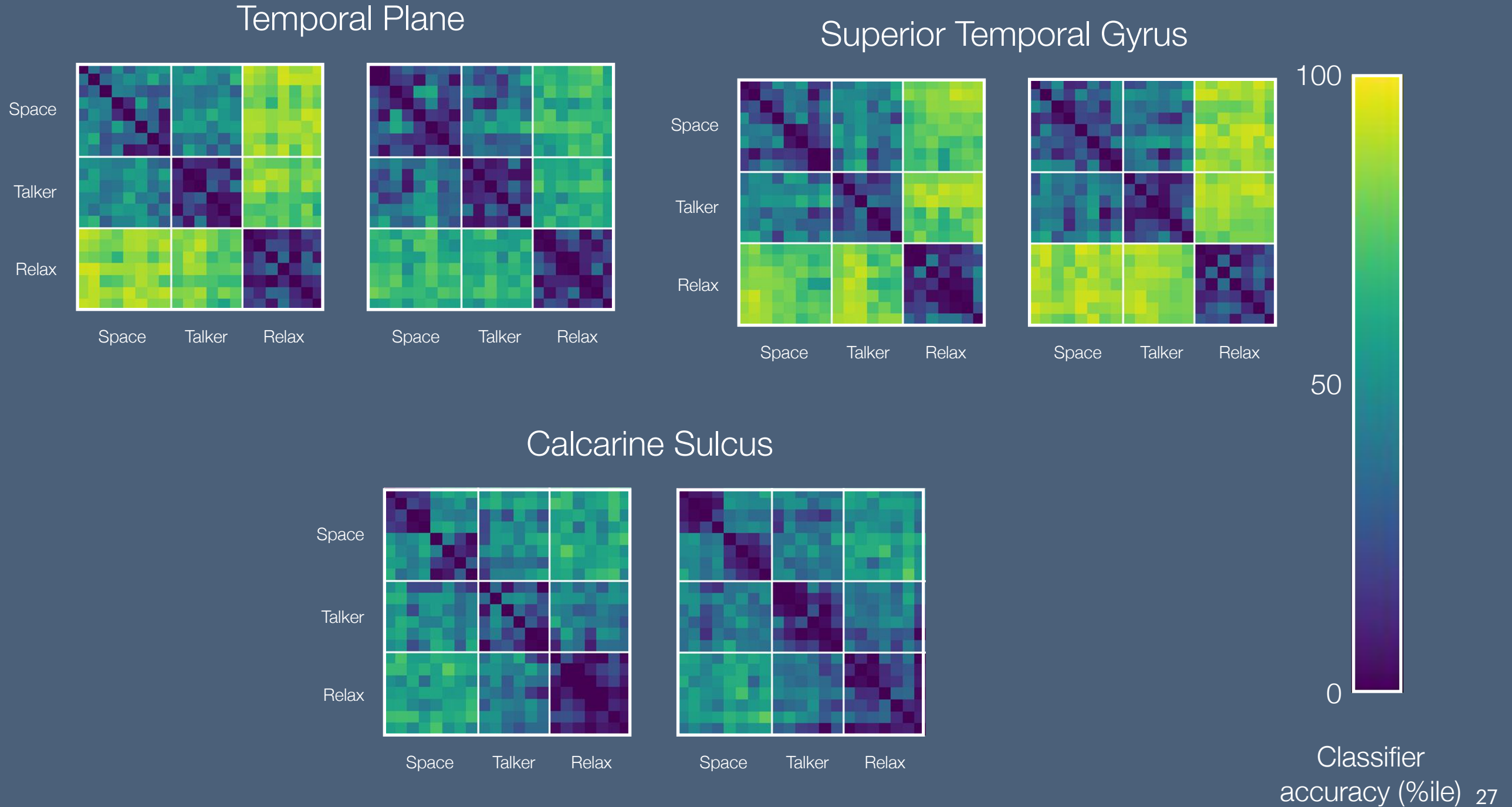
Representational similarity using searchlight MVPA



Strong attention vs. passive information, but
weak spatial vs. talker attention discrimination

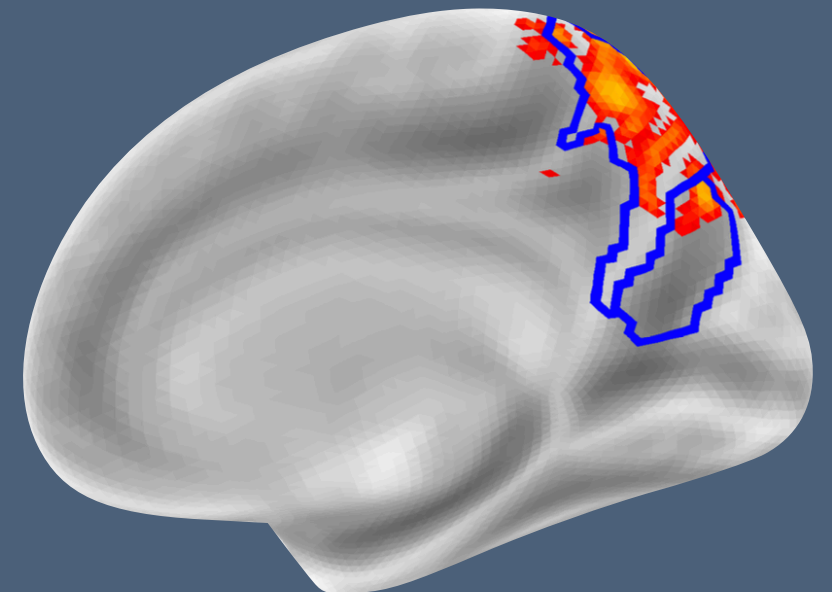
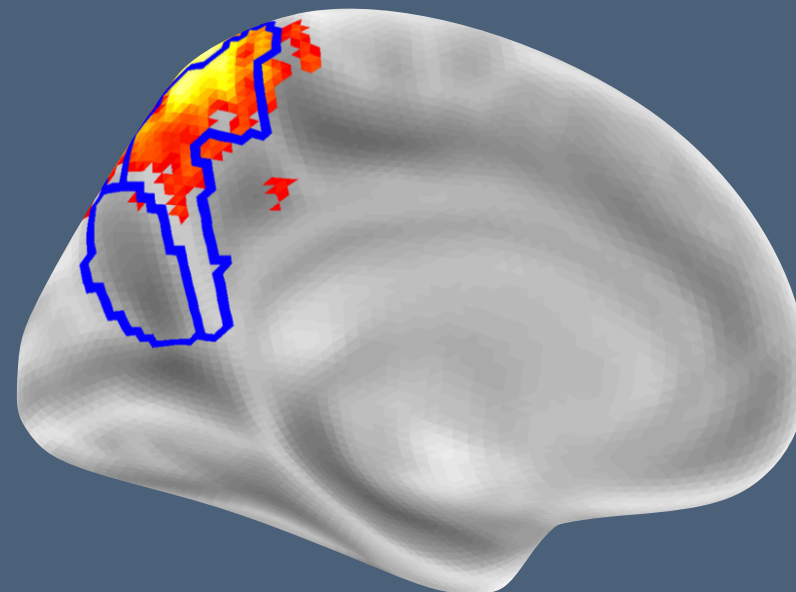
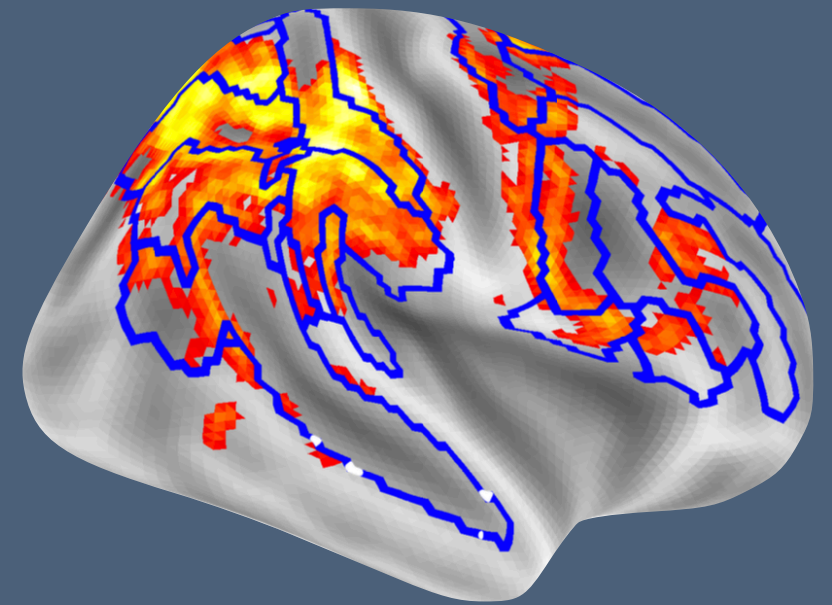
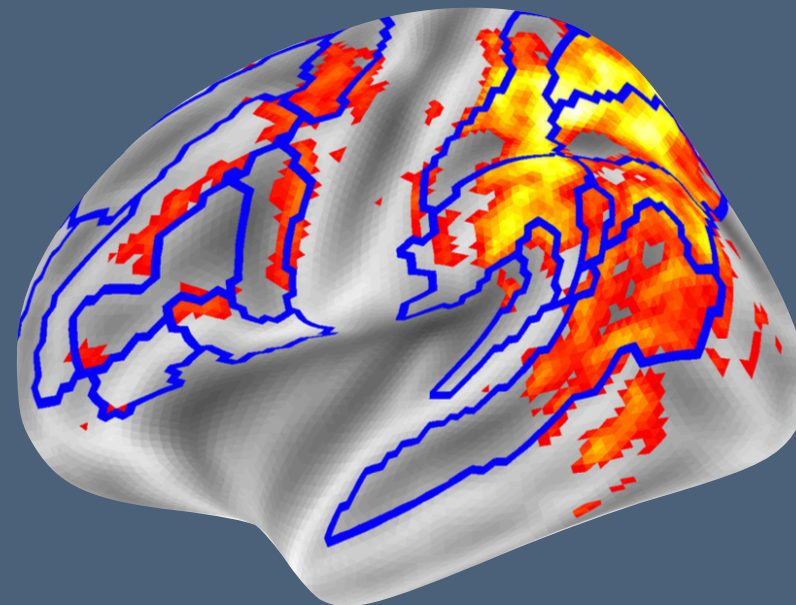


Early visual cortex is less decodable for attention vs passive listening... but carries **location**



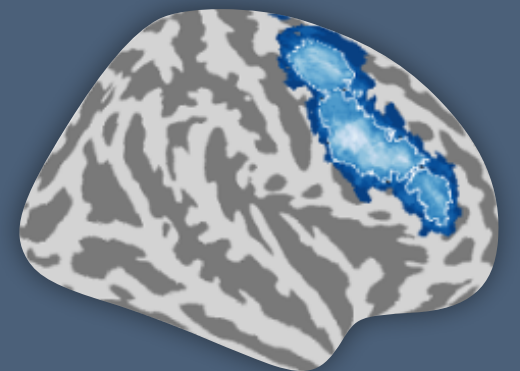
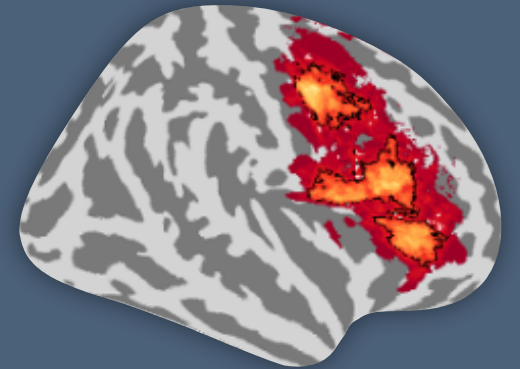
Auditory attention across the brain

- Representations of attention type (spatial vs. talker)
- parietal
- frontal
- posterior temporal



Conclusions

- PFC contains discrete networks specialized to perform computations needed for hearing and vision.
- Networks support the “wrong” modality when their computational domain is needed:
 - Dual-task interference
 - Spatial vs. non-spatial memory
 - Representational geometry of attention
- Future: **When and how do PFC networks communicate and work together?**



All the references!

Thanks to



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